

MOEN-IMS: Inventory Management System

Ministry of Energy — Integrated Energy Project System (IEPS)

System Overview Report

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1. Purpose

MOEN-IMS is a web-based platform that manages and tracks construction and project materials across their full lifecycle — from warehouse stock through requisition, approval, transport, and confirmed delivery to site. It replaces paper-based forms, telephone coordination, and disconnected spreadsheets with a single system that gives every stakeholder a shared, real-time view of material movements and an auditable record of every transaction.

2. System at a Glance

Attribute	Detail
Type	Web application (browser-based, mobile-friendly)
Core function	End-to-end material request, tracking and delivery management
User roles	6 (Administrator, Schedule Officer, Storekeeper, Management, Consultant, Transporter)
Deployment	Cloud-hosted; supports concurrent multi-office use
Status	Operational, with an enhanced feature set in phased rollout

3. Technology and Architecture

The system is built on the Django (Python) web framework using a Model–View–Template architecture, with a PostgreSQL database in production. The application follows a signal-based design so that notifications, status updates, and audit entries are generated automatically as records change.

- **Backend:** Django 5.1 (Python), served via Gunicorn.
- **Database:** PostgreSQL (SQLite for development).
- **Frontend:** HTML, CSS/Bootstrap, and JavaScript; responsive on desktop and mobile.
- **Data & reporting:** pandas and openpyxl for Excel import/export; ReportLab, Plotly and Matplotlib for PDF reports and dashboards.
- **Supporting services:** QR-code generation and verification, Microsoft 365 e-mail notifications, and one-time-password (OTP) two-factor authentication.

4. User Roles

Access is role-based; each user sees only the functions relevant to their responsibilities.

5. Core Capabilities

- **Material request workflow:** digital requisitions with unique tracking codes moving through a Draft → Pending → Approved → Completed lifecycle.
- **Inventory dashboard:** real-time stock levels with colour-coded low-stock alerts.

Role	Primary Responsibility
Administrator	Full system access: user management, configuration, and oversight of all records.
Schedule Officer	Raise material requests for projects and track their status.
Storekeeper	Process requests, manage inventory levels, and assign transporters.
Management	Oversight, approvals, reporting, and analytics across all activity.
Consultant	Confirm site deliveries, log receipts, and report discrepancies.
Transporter	Execute deliveries and update transport status to completion.

- **Transport management:** approved transporter and vehicle records, with status tracking from Assigned → In Transit → Delivered.
- **Waybill generation:** automatic, uniquely numbered waybills carrying QR codes for on-the-spot verification.
- **Budget (BOQ) control:** requests checked against Bill of Quantities allocations, with automatic detection and controlled approval of over-issuance.
- **Site receipt confirmation:** consultants log delivery, condition, and any discrepancies as digital proof of receipt.
- **Reporting:** automated weekly and management reports exported to PDF and Excel.
- **Bulk operations:** bulk transporter assignment and consignment tracking for multi-order deliveries.

6. Security and Controls

The platform applies layered, industry-standard controls to protect data and ensure accountability:

- Role-based access control and session management.
- Two-factor authentication (OTP) and strong password hashing.
- Protection against SQL injection, cross-site scripting (XSS), cross-site request forgery (CSRF), and clickjacking.
- Encrypted transport (SSL/HTTPS) and validated file uploads.
- QR-verified waybills and cryptographic digital signature stamps to deter fraud and forgery.
- Complete audit trail of user actions for regulatory compliance.

7. Current Status and Outlook

The core system is operational in production, with an enhanced feature set — including assigned-order stores management, QR digital waybills, digital signature stamps, BOQ over-issuance controls, performance dashboards, and automated reporting — being deployed in phases. The cloud-based architecture is designed to scale with demand, and planned next steps include an offline mode for low-connectivity sites and API integration with accounting systems.

Prepared as a system overview of the MOEN-IMS platform. Internal use only.